THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHEMATICS TWO-CHAPTER TEST SET B

Chapters 1 & 2: **Real Numbers, Polynomials** | RBSE Board Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The prime factorisation of 432 is: 1

(a) $2^3 \times 3^4$ **(b)** $2^4 \times 3^2$ **(c)** $2^2 \times 3^3$ **(d)** $2^4 \times 3^3$

2. The LCM of $2^3 \times 3$ and $2 \times 3 \times 5$ is: 1

(a) 120 **(b)** 6 **(c)** 240 **(d)** 30

3. Which of the following is irrational? 1

(a) $\frac{\sqrt{12}}{\sqrt{3}}$ **(b)** $\sqrt{\frac{4}{9}}$ **(c)** π **(d)** 0.5

4. A student states: "The HCF of two numbers can be greater than their LCM." The correct response is: 1

(a) Correct for primes **(b)** Incorrect — HCF ≤ LCM always
(c) Correct **(d)** Cannot be determined

5. For HCF(18, 24), Student A gets 6 and Student B gets 72. Then: 1

(a) Student A is correct **(b)** Student B is correct
(c) Both are correct **(d)** Both are wrong

6. Which is a zero of $x^2 - 3x + 2$? 1

(a) 3 **(b)** −2 **(c)** −1 **(d)** 2

7. If the sum of the zeroes of $kx^2 - 3x + 5$ is 1, then k is: 1

(a) 5 **(b)** 3 **(c)** −3 **(d)** $\frac{1}{3}$

8. The graph of $y = ax^2 + bx + c$ with $a < 0$ is a parabola that opens: 1

(a) to the right **(b)** to the left **(c)** downwards **(d)** upwards

9. The zeroes of $x^2 + 4x - 21$ are: 1

(a) $-21, 1$ **(b)** $-3, 7$ **(c)** $7, 3$ **(d)** $3, -7$

10. A quadratic polynomial has sum of zeroes 0 and product -5. Student A writes $x^2 - 5$; Student B writes $x^2 + 5$. Then: 1

(a) Student B is correct **(b)** Student A is correct
(c) Both are correct **(d)** Both are wrong

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the HCF and LCM of 40, 36 and 126 by prime factorisation. 2

12. Can 12^n end with the digit 0 or 5 for any natural number n? Justify. 2

13. Find a quadratic polynomial whose zeroes are $2+\sqrt{3}$ and $2-\sqrt{3}$. 2

14. If α, β are the zeroes of $3x^2+5x-2$, find $\alpha^2+\beta^2$. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Prove that $7-2\sqrt{3}$ is irrational, given that $\sqrt{3}$ is irrational. 3

16. Find the zeroes of $x^2+\frac{1}{6}x-2$ and verify the relationship between the zeroes and the coefficients. 3

17. Find the least number which when divided by 15, 18 and 24 leaves remainder 4 in each case. 3

OR

Find the largest number that divides 2053 and 967 leaving remainders 5 and 7 respectively. 3

18. If α, β are the zeroes of $x^2-p(x+1)-c$, show that $(\alpha+1)(\beta+1)=1-c$. 3

OR

If -3 is a zero of x^2+2x+k, find k and the other zero. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study:** A rectangular field 120 m × 84 m is to be fenced with poles placed at equal distances along its boundary, with a pole at each corner. 5

(i) Find the maximum possible distance between two poles.
(ii) Find the number of poles.
(iii) How many poles are needed if the distance is 6 m?

OR

(i) Find the HCF and LCM of 1176 and 1260 by prime factorisation and verify HCF $\times$ LCM $=$ product. 5
(ii) Check that the LCM is divisible by both numbers.

20. **Case study:** A diving bird's path relative to the water surface is $h(x)=x^2-6x+8$ (in m), where $h<0$ means below water. 5

(i) Find the points where it enters and leaves the water.
(ii) How deep is it at $x=3$?
(iii) Verify the relationship between the zeroes and coefficients.

OR

Verify that $2, -1$ and 4 are the zeroes of x^3-5x^2+2x+8, and verify all three relationships between the zeroes and the coefficients. 5

END OF QUESTION PAPER — **Total: 40 Marks**

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